

MLFF EVAL-MF1/MF2 multi-fidelity checkpoint-evaluation specification

Status: EVAL-MF1 implemented in mdstats 0.20.106a0; EVAL-MF2 implemented in mdstats 0.20.107a0. PREC1 is implemented in mdstats 0.20.108a0; PREC2 is the next MLFF implementation gate.

Purpose

This specification defines the production replacement for the historical four-checkpoint screening default. The evaluator spends a small, equal fraction of target and replay monitor labels on every saved epoch, progressively increases fidelity for survivors, and performs final scientific selection only on complete authoritative monitor data.

The implementation is split into two gates:

1. **EVAL-MF1** - deterministic nested monitor ladders, incremental prediction reuse, multi-round execution, full-fidelity finalist evaluation, and durable restart records.
2. **EVAL-MF2** - conservative statistical guard bands, ambiguity-driven expansion, comprehensive epoch reporting, exhaustive-comparison qualification, and migration to the production default.

EVAL-MF1 implementation record

mdstats 0.20.106a0 implements the first gate as an opt-in campaign strategy. The runtime uses the original authenticated complete monitor artifacts plus immutable configuration-index ladders; it does not manufacture alternate mini-monitor source files. Candidate predictions are persisted as immutable coverage shards keyed by model/checkpoint identity, head, numerical contract, and ordered geometry identities. A later cumulative round composes valid earlier shards and evaluates only missing geometry identities. Dedicated partial-round records carry explicit fidelity/evidence class and survivor records persist deterministic rank/outcome/reason data. Only the 100% round publishes the ordinary authoritative checkpoint-evaluation record.

The 0.20.106a0 EVAL-MF1 survivor rule was deliberately deterministic and simple. EVAL-MF2 in 0.20.107a0 implements the statistical ambiguity/guard-band behavior below and advances newly generated campaign TOML to `checkpoint_strategy = "multi_fidelity"`. Legacy configs that omit the strategy continue to resolve to bounded for restart compatibility.

EVAL-MF2 implementation record

mdstats 0.20.107a0 implements conservative source/temporal-block survivor evidence and comprehensive epoch reporting on top of EVAL-MF1. The default guard retains a candidate when the paired mean primary-metric difference from the nominal cutoff is no larger than a 2% relative allowance plus 2 standard errors over at least 4 common blocks. If fewer blocks are available, only the deterministic 2% relative allowance is used. True-label replay campaigns reserve provisionally replay-compatible candidates up to the minimum- finalist floor, and a pairwise rank-inversion fraction $\geq 25\%$ expands the subsequent round to at least 50% of current candidates. These numbers are explicit policy fields, not hidden constants.

Every run receives normative JSON plus CSV/Markdown epoch-evaluation reports that combine MACE training history with partial/full independent metrics, survivor reasons, final admissibility, and selected status. A representative deterministic 30-checkpoint qualification selects the same true-replay-admissible winner as exhaustive evaluation while purchasing 10.89 rather than 30 full-checkpoint-equivalent candidate inference, a 63.7% reduction for that case. Supplied MACE 0.3.16 restoration and monitor-graph cache regressions separately qualify the real-model inference substrate.

Binding invariants

1. Every saved candidate checkpoint enters round 1 unless it is already invalid by immutable checkpoint/catalog evidence (for example corrupt bytes).
2. A round fraction f applies to the same nominal fraction of the complete target monitor and the complete true-label replay monitor. Neither domain is intentionally assigned a higher evaluation fidelity.

3. Target and replay metrics remain separate; the implementation shall not pool them into a single RMSE merely for screening.
4. Subsets are deterministic, label-independent, stratified, and nested.
5. Later rounds infer only the delta not already authenticated in earlier rounds.
6. Partial-round metrics are screening evidence only.
7. Final checkpoint admissibility and selection require complete target data and, for replay-backed protocols, complete true-label replay data.
8. The existing frozen checkpoint metric/admissibility/selection policy remains authoritative in the full-fidelity round.
9. An incomplete or partially evaluated checkpoint is never represented as a scientifically rejected checkpoint.
10. Evaluation artifacts are campaign-owned; source monitor files remain read-only.

Default ladder

The initial policy to qualify is:

Round	Target fraction	Replay fraction	Nominal survivor action
1	0.10	0.10	EVAL-MF1: retain the nominal best one third subject to the minimum-finalist floor; EVAL-MF2 adds guard bands
2	0.33	0.33	EVAL-MF1: retain at least the configured finalist floor; EVAL-MF2 adds guard bands
3	1.00	1.00	full admissibility and deterministic selection

Fractions and survivor controls are configurable execution policy. Integer and stratum-minimum rounding may produce slightly different realized counts while preserving the same nominal fraction on both monitor domains.

Nested subset construction

For each monitor domain, build one immutable ordered configuration catalog before candidate results are inspected. The order shall balance all frozen monitor strata available to the evaluator, including condition axes and source/trajectory grouping. Within a trajectory/source, temporally nearby frames shall not be treated as iid samples when a decorrelated or block-aware ordering can be constructed from existing partition evidence.

Round subsets are prefixes of the immutable order. Their identities bind the complete monitor digest, ordered configuration identities, stratum policy, round fractions, and realized sizes.

Prediction persistence

Candidate prediction storage shall support authenticated incremental coverage. A surviving checkpoint evaluated on round $r+1$ predicts only configurations absent from round r . Metric reduction reads the union of authenticated

prediction shards. A direct one-shot evaluation on the same complete subset must be numerically identical within the existing precision tolerance.

Existing OPT-EVAL2 foundation prediction reuse and OPT-EVAL3 monitor graph/view caches remain authoritative when identities match.

Survivor ranking

EVAL-MF1 may use deterministic provisional ranking derived from the existing frozen checkpoint policy, but partial threshold outcomes are not final admissibility records. A nominal one-third survivor fraction is a resource target, not a hard scientific cap.

EVAL-MF2 implements paired, block-aware ambiguity handling. Because checkpoints are evaluated on the same configurations, comparisons should use paired loss/error differences where possible. Uncertainty estimation shall operate on trajectory/source or decorrelated blocks rather than treating individual adjacent MD frames as independent samples. Candidates statistically indistinguishable from a cutoff are retained. A detected boundary-ranking inversion triggers deterministic expansion rather than forced pruning.

The production policy shall not claim a global-minimum guarantee over all saved epochs unless all epochs receive full-monitor evaluation. It guarantees exact application of the frozen selection policy among the full-fidelity finalists retained by the declared screening policy.

Restart records

Every round persists:

- round plan and subset identities;
- per-checkpoint authenticated prediction coverage;
- partial metrics with explicit fidelity tags;
- survivor/elimination ranking inputs and reasons;
- remaining inference deltas;
- final full-fidelity metric/admissibility records for finalists.

Restart shall reuse every valid prediction shard and recompute only missing/corrupt deltas.

Comprehensive epoch report

EVAL-MF2 produces a machine-readable and human-readable summary for every epoch, including when available:

- MACE training-history target/replay metrics;
- independent target energy and force errors by round;
- focus-group and worst-condition metrics;
- replay candidate/baseline metrics and degradation;
- round fraction and realized monitor counts;
- survivor/elimination reason;
- final admissibility reason; and
- selected/nonselected status.

JSON is normative. CSV and plotting-ready outputs are convenience derivatives.

Qualification

The production-default qualification completed in 0.20.107a0 demonstrates:

1. deterministic nested equal-fraction target/replay subsets;
2. label-independent subset construction;

3. all saved epochs in round 1;
4. exact incremental prediction reuse;
5. restart/corruption recovery;
6. no final scientific decision from partial metrics;
7. full target/replay evaluation of every finalist;
8. numerical identity against one-shot full evaluation of the same finalist;
9. conservative behavior under synthetic ranking inversions/noisy early subsets; and
10. representative comparison with exhaustive 30-checkpoint evaluation, recording both winner agreement and inference-cost reduction.

The current fixed `max_checkpoints_per_run` mode remains available as an explicit fast or compatibility strategy. Exhaustive full-monitor evaluation remains available as an audit/reference strategy.

0.20.107a0 qualification evidence

- representative exhaustive comparison: 30 saved checkpoints, exhaustive winner epoch 16, multi-fidelity winner epoch 16;
- nested candidate counts in that case: 30 -> 11 -> 8;
- incremental candidate-inference cost: 10.89 full-checkpoint equivalents versus 30 exhaustive, or 63.7% less candidate inference;
- real-MACE qualification: supplied MACE 0.3.16 direct checkpoint restoration and persistent monitor-graph/cache regressions pass; and
- partial screening remains non-authoritative and acquires no checkpoint-deletion authority.

The measured reduction is case-specific and is not a universal performance guarantee.